Enrollment No._____

GANPAT UNIVERSITY

M.Sc.IT(CS) Sem-III Regular Nov-Dec 2024

P73A1MTA : MALWARE TAXONOMY AND ANALYSIS

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Define Malware Analysis and Explain different techniques used to study Malware?	06	CO1	BTL4
2	Discuss goals for Malware Analysis and write down thumb rules for Malware Analysis?	06	CO1	BTL5
3	What is Snapshot? Explain the Pros and Cons of taking Snapshot during Malware Analysis Process ?	06	CO3	BTL3
4	Explain Antivirus Scanning and Hashing and its importance for Malware Analysis.	06	CO2	BTL4
5	Explain how Procmon (Process Monitor) is used in malware analysis. What specific types of activities can it help monitor, and why is this important?	06	CO4	BTL5
6	Write a short note on Persistence Mechanism.	06	CO5	BTL3
7	Write down steps to create your virtual machine for Malware Analysis	06	CO3	BTL6

Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Explain the role of strings in malware analysis and how they can help identify malicious activity.	06	CO2	BTL3
2	Define Malware and Discuss the various types, such as Trojan, Ransomware, Spyware, Downloader, Worm, and Adware and explain what each type does.	06	CO1	BTL2
3	What are packed and obfuscated malware, and how do these techniques help malware evade detection?	06	CO2	BTL4

Q.3	**Answer the following: (Any Two)**	**12**		
1	Describe the structure of a virtual machine and discuss the advantages of using virtual machines for malware analysis compared to physical machines.	06	CO3	BTL6
2	Write a short note on Wireshark and Inetsim.	06	CO4	BTL3
3	What is sandboxing in malware analysis, and what are some of its potential drawbacks?	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain Backdoors and Write a short note on Reverse Shell.	06	CO5	BTL3
2	Discuss how credential stealers operate within malware. What methods do they use to capture sensitive information, and what can be done to mitigate these risks?	06	CO5	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

M.Sc. IT (Cyber Security) Sem-III Regular/Remedial Nov-Dec 2024

P73A2MDS : MOBILE DEVICE SECURITY AND FORENSIC

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	How does android o/s work? Describe its architecture.	06	1	BTL-4
2	Demonstrate any three controls of android with example.	06	1	BTL-5
3	Write a short note on Burp Suite.	06	2	BTL-3
4	Explain Android Startup Process with steps.	06	2	BTL-5
5	Describe the vulnerability of content provider leakage.	06	3	BTL-4
6	Write a short note on packet capture.	06	4	BTL-3
7	How to logging with Logcat?	06	5	BTL-6
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Describe Activity and its lifecycle.	06	1	BTL-4
2	What is R.java? Describe the resources in detail.	06	1	BTL-3
3	What is ADB? Explain its commands.	06	2	BTL-5
Q.3	**Answer the following: (Any Two)**	**12**		
1	Define reverse engineering. How to do it for android app?	06	3	BTL-5
2	Describe the passive analysis in detail.	06	4	BTL-4
` 3	What is proxy interception? Describe the steps in detail.	06	4	BTL-4
Q.4	**Answer the following: (Any One)**	**06**		
1	Elaborate the types of forensic.	06	5	BTL-4
2	Write a short note on the android filesystem partition.	06	5	BTL-4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

M.Sc.IT (Cyber Security) Sem-III Regular Nov-Dec 2024

P73A3NF: NETWORK FORENSIC

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Justify the classification of network forensic systems.	06	CO1	BTL5
2	Describe the network forensic process models.	06	CO2	BTL1
3	Explain the Nessus in detail.	06	CO3	BTL2
4	Point out the challenges in network forensic analysis.	06	CO1	BTL4
5	Apply the concept of OmniPeek tool.	06	CO3	BTL3
6	Identify the Network Miner in brief.	06	CO4	BTL1
7	Specify the TCP/IP protocol suite in detail.	06	CO5	BTL6
Section – II				
Q-2	**Answer the following. (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Explain the Honeypot forensics with example.	06	CO1	BTL6
2	Discuss the data mining-based frameworks in detail.	06	CO2	BTL2
3	Define the Botnet forensics in brief.	06	CO1	BTL1
Q.3	**Answer the following. (Any Two)**	**12**		
1	Discover the concept of WebScarab tool.	06	CO4	BTL3
2	Analyze IDS Snort in detail.	06	CO4	BTL4
3	Justify the Xplico tool in brief.	06	CO3	BTL5
Q.4	**Answer the following. (Any One)**	**06**		
1	Justify the packet capture format in detail.	06	CO5	BTL5
2	Discuss the NetFlow record format in brief.	06	CO5	BTL2

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

M.Sc. IT (CS) Sem- III Regularl Nov-Dec 2024

P73B5ISC: INFORMATION SECURITY COMPLIANCE MANAGEMENT

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Describe the main principles of Information System Security and their importance.	06	CO1	BTL2
2	Explain the ISO/IEC 27001 specification and its key components.	06	CO3	BTL2
3	What are the major components involved in organizing Information Security in a business?	06	CO4	BTL3
4	Discuss the purpose and rules of HIPAA with respect to information security.	06	CO2	BTL4
5	Explain the different types of IPR.	06	CO4	BTL1
6	What are the benefits of implementing ISMS?	06	CO4	BTL4
7	Discuss the ISO 27001 certification process.	06	CO2	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Discuss the need for Information Security and the benefits it brings to an organization.	06	CO2	BTL4
2	Explain the risk management process within the ISMS framework under ISO 27001.	06	CO1	BTL4
3	Discuss different clauses in ISO/IEC 27001.	06	CO4	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain asset management in detail.	06	CO2	BTL3
2	Explain PCI-DSS in brief.	06	CO3	BTL2
3	What are the organizational responsibilities for maintaining ISMS standards?	06	CO2	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Discuss the importance of ethics in Information Security.	06	CO2	BTL5
2	How does IPR differ between India and other countries, and why is it important for businesses?	06	CO3	BTL4

**** End of the Paper ****